



Lab Assignment – 4 (JOIN, WITH, LATERAL, STRING & CASE OPERATIONS)

Department: CSE

Subject: DIS Lab
Code: 22CPP217

Objective: Understand data distribution across multiple tables and implement INNER JOIN, WITH (CTE), LATERAL, STRING functions, and CASE statements to solve real-world database problems using conditional logic.

Reference :

Pre-Lab Setup:

SQL

```
-- 0. Disable Safe Update Mode & reset your tables
```

```
SET SQL_SAFE_UPDATES = 0;
```

```
DROP TABLE IF EXISTS table_name;
```

```
-- 2. Create Schema
```

```
Student - Student_ID(primary key),Name (not null),Dept_Name
```

```
Course - Course_ID(primary key), Course_Name, Dept_Name ,Credits
```

```
Enrollment - Enroll_ID(primary key), Student_ID, Course_ID, Semester, Grade()
```

```
-- 3. Insert Data into Tables
```

```
Student
```

```
(1, ' aaRAV shaRMa ', 'CSE'), (2, 'neha VERMA', 'CSE'),  
(3, 'Rohan Singh', 'Physics'), (4, 'Meera Joshi', 'Math'),  
(5, 'Ishita Roy', 'History')
```

```
Course
```

```
(101, 'DBMS', 'CSE', 4), (102, 'Operating Systems', 'CSE', 3),  
(103, 'Quantum Mechanics', 'Physics', 4), (104, 'Linear Algebra', 'Math', 3)
```

```
Enrollment
```

```
(1, 1, 101, 'Sem1', 'A'), (2, 1, 102, 'Sem1', 'B'),  
(3, 2, 101, 'Sem1', 'A'), (4, 3, 103, 'Sem1', 'C'),  
(5, 4, 104, 'Sem1', 'A'), (6, 2, 102, 'Sem1', 'B')
```

Q 1: The university needs to link student records with their academic performance(Natural join).

- Write a query to display the **Student Name**, **Course Name**, and **Grade** for all enrolled students.
 - Use INNER JOIN to link Student, Enrollment, and Course tables.
- List the names of students who are currently enrolled in any course offered by the '**CSE**' department.
- Retrieve the names of students and the course names where the student achieved a Grade of '**A**'.

Q 2: The data in the Student table was entered manually and contains errors (extra spaces, weird capitalization). You must fix this for the official report.(string operations)

- Display the Student_ID and Name. The name must be formatted such that:
 - Extra spaces are removed from the start and end (TRIM).
 - The name is converted to UPPERCASE (UPPER).
- Concatenate the **first 3 letters** of the Student's Name with their **ID** to create a unique login ID (e.g., AAR1 or NEH2).
- Display students whose name starts with the letter '**A**' or '**M**' (Use LIKE or LEFT).

Q 3: The exam board wants to categorize performance without changing the data.(case)

- Display Student Name, Grade, and a new column '**Remarks**' based on:
 - Grade = 'A' → '**Excellent**'
 - Grade = 'B' → '**Good**'
 - Else → '**Needs Improvement**'
- Display Course Name and categorize it as:
 - Credits ≥ 4 → '**Core Course**'
 - Else → '**Elective Course**'

Q 4: Use Common Table Expressions to simplify complex calculations.(with/cte)

- Create a CTE named Student_Credits that computes the **total credits** taken by each student. Use this CTE to display only those students who have taken **more than 5 credits**.
- Create a CTE that counts how many students are enrolled in each department. Display the departments with more than 1 enrollment.

Q 5: Advanced Row-Dependent Queries (lateral) (check: MySQL version 8.0.14+)

- For every student, retrieve the **one** course where they achieved their highest grade.
 - Select from Student table, then JOIN LATERAL with a subquery that sorts Enrollment by Grade (ORDER BY Grade ASC) for that specific student and uses LIMIT 1.

SQL

-- syntax hints

-- inner join

```
SELECT t1.column_name, t2.column_name
FROM Table1 t1
JOIN Table2 t2 ON t1.common_id = t2.common_id;
```

-- string functions

TRIM(column_name) removes leading/trailing spaces.
UPPER(column_name) converts text to uppercase.
LEFT(column_name, 3) extracts the first 3 characters.

-- case

```
CASE
    WHEN column_name = 'Value1' THEN 'Output A'
    WHEN column_name >= 10 THEN 'Output B'
    ELSE 'Default Output'
END AS New_Column_Label
```

-- CTE(with)

```
WITH CTE_Name AS (
    SELECT column_name, COUNT(*) as Total
    FROM Table_Name
    GROUP BY column_name
)
SELECT * FROM CTE_Name WHERE Total > 5;
```

-- Lateral derived table

```
SELECT t1.name, sub.result
FROM Table1 t1,
LATERAL (
    SELECT col_name
    FROM Table2 t2
    WHERE t2.id = t1.id
    ORDER BY some_col DESC LIMIT 1
) sub;
```